JEE APEX FTS – JEE ADV. PART TEST-01 [P2]

KEY SHEET

MATHEMATICS

1	**BC**	2	**BCD**	3	**ABCD**	4	**B**	5	**C**	6	**B**
7	**C**	8	**10**	9	**3**	10	**13**	11	**24**	12	**8**
13	**5**	14	**C**	15	**D**	16	**A**	17	**B**		

PHYSICS

18	**ABC**	19	**ABCD**	20	**BC**	21	**C**	22	**A**	23	**B**
24	**D**	25	**5**	26	**4**	27	**8**	28	**3**	29	**8**
30	**4**	31	**A**	32	**B**	33	**A**	34	**B**		

CHEMISTRY

35	**BC**	36	**CD**	37	**AD**	38	**C**	39	**A**	40	**B**
41	**B**	42	**7**	43	**7**	44	**9**	45	**5**	46	**3**
47	**6**	48	**A**	49	**B**	50	**C**	51	**D**		

SOLUTIONS

MATHEMATICS

1. $\det\left(A^2-2I\right)=0 \Rightarrow \det\left(A-\sqrt{2}I\right)\det\left(A+\sqrt{2}I\right)=0$

As A contains rational entries, $\det\left(A-\sqrt{2}I\right)=\det\left(A+\sqrt{2}I\right)=0$

So characteristics equation of A is $A^2 - 2I_2 = 0$

$A^{2n} = 2^n I_2, A^{2n+1} = 2^n A \& |A| = -2.$

2. As sum of coefficients of $x^2, y^2 \text{ and } z^2$ is zero so the planes are perpendicular

Clearly $x = y = z$ satisfies the equation. So $l = m = n = \frac{1}{\sqrt{3}}.$

3. If $OA = x, OB = y, OC = z.$ $p = q = \frac{yz}{\sqrt{y^2 + z^2}}. \lambda = \frac{z^2}{y^2} \& \mu = \frac{y^2}{z^2}.$

4. $|P(0)| \le 0 \Rightarrow a_0 = 0$

Now $|a_1 x + a_2 x^2 + ...| \le |x|$ $|a_1 + a_2 x + ...| \le 1$ So $|a_1| \le 1.$

5. Maximum value of A is 398.

6. $a_2 = 2$ so $a_n > 2 \forall n > 2.$ $a_n^2 = a_{n-1}^2 + \frac{1}{a_{n-1}^2} + 2$

As $a_n > 2$ so $\frac{1}{a_n^2} < \frac{1}{4} < 1$

Hence $2 < a_n^2 - a_{n-1}^2 < 3$

putting $n = 2,3,...,75$ and adding $148 < a_{75}^2 - 1 < 222$

$144 < 149 < a_{75}^2 < 223 < 225$ So $12 < a_{75} < 15$

7. Sum (required) =

$$\sum_{i=1}^{12} \left[(i-1)12 + i \right] = 12(0 + 1 + ... + 11) + (1 + 2 + 3 + ... + 12) = 870$$

8. Note that the left-hand side of the equation can be written as

$x(x^2 - 2m + 1) + m(x^2 - 2m + 1) + x^2 - 2m + 1 = 5$

Then, $(x + m + 1)(x^2 - 2m + 1) = 5$

Now, if the equation has an integer root, we have

$(x + m + 1, x^2 - 2m + 1) \in \{(1,5), (-1,-5), (5,1), (-5,-1)\}$

We have four cases,

(i) $x + m + 1 = 1 \text{ and } x^2 - 2m + 1 = 5.$

Then, $x^2 - 2m + 1 + 2(x + m + 1) = 7.$

Hence, $x^2 + 2x - 4 = 0,$ which has no integer roots.

(ii) $x + m + 1 = -1 \text{ and } x^2 - 2m + 1 = -5.$ Then, $x^2 + 2x + 10 = 0,$ which has no real roots.

(iii) $x + m + 1 = 5 \text{ and } x^2 - 2m + 1 = 1.$ Then $x^2 + 2x - 8 = 0,$ which gives $x = 2, -4 \text{ and }, m = 2,8$ respectively.

(iv) $x+m+1=-5 \text{ and } x^2-2m+1=-1$. Then $x^2+2x+14=0$, which has no real roots.

9. $a=2, b=6, c=3, d=4$

10. $\sin\left(1+\sin^2 x+\cos^4 x\right)=\dfrac{13}{14}$

$\cos\left(1+\sin^2 x+\cos^4 x\right)=\dfrac{-3\sqrt{3}}{14}$

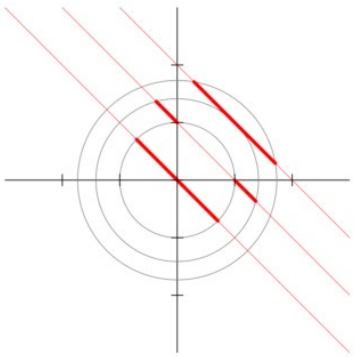
11. We split into cases on the integer $k=\left[x^2+y^2\right]$. Note that $x+y=k$ but $x^2+y^2 \geq \dfrac{1}{2}(x+y)^2=\dfrac{1}{2}k^2$ and $x^2+y^2<1$, which forces $k \leq 2$.

If $k=0$, the region defined by $0 \leq x^2+y^2<1$ and $x+y=0$ is the diameter from $\left(\dfrac{\sqrt{2}}{2},-\dfrac{\sqrt{2}}{2}\right)$ to $\left(-\dfrac{\sqrt{2}}{2},\dfrac{\sqrt{2}}{2}\right)$, which has length 2.

If $k=1$, the region $1 \leq x^2+y^2<2$ and $x+y=1$ consists of two segments, which is the chord on $x^2+y^2=2$ minus the chord on $x^2+y^2=1$. The former has length $2\sqrt{\left(\sqrt{2}\right)^2-\left(\dfrac{\sqrt{2}}{2}\right)^2}=\sqrt{6}$, and the latter has length $2\sqrt{1^2-\left(\dfrac{\sqrt{2}}{2}\right)^2}=\sqrt{2}$. So the total length here is $\sqrt{6}-\sqrt{2}$.

If $k=2$, the region $2 \leq x^2+y^2<3$ and $x+y=1$ is the chord on $x^2+y^2=3$, which has length $2\sqrt{\left(\sqrt{3}\right)^2-\left(\sqrt{2}\right)^2}=2$.

Our final answer is $2+\left(\sqrt{6}-\sqrt{2}\right)+2=4+\sqrt{6}-\sqrt{2}$.



12. $H(n)=5\left(1+2+3+....+\left[\dfrac{n}{5}\right]\right)+5^2\left(1+2+3+....+\left[\dfrac{n}{5^2}\right]\right)+....$ so. On. ([.] G.I.F)

Using sandwich theorem, we can show $\lim\limits_{n\to\infty}\dfrac{n^2}{H(n)}=8$.

13. The given condition is equivalent to $b=\dfrac{a+c}{1-ac}$.

$a=\tan\alpha,\ b=\tan\beta,\ c=\tan\gamma,$

where $\tan\beta=\tan(\alpha+\gamma)\ and\ \alpha,\beta,\gamma\in\left(-\dfrac{\pi}{2},\dfrac{\pi}{2}\right)$, so we have

$$A=\frac{2}{a^2+1}-\frac{2}{b^2+1}+\frac{3}{c^2+1}=\frac{2}{\tan^2\alpha+1}-\frac{2}{\tan^2(\alpha+\gamma)+1}+\frac{3}{\tan^2\gamma+1}$$

$$=2\cos^2\alpha-2\cos^2(\alpha+\gamma)+3\cos^2\gamma$$

$$=(2\cos^2\alpha-1)-(2\cos^2(\alpha+\gamma)-1)+3\cos^2\gamma$$

$$=\cos\alpha-\cos(2\alpha+2\gamma)+3\cos^2\gamma\ =2\sin(2\alpha+\gamma)\sin\gamma+3\cos^2\gamma$$

Let $x=|\sin\gamma|$. Then we have

$$A\le 2x+3(1-x^2)=-3x^2+2x+3=-3\left(x-\frac{1}{3}\right)^2+\frac{10}{3}\le\frac{10}{3}.$$

Equality holds if any only if $\sin(2\alpha+\gamma)=1$ and $\sin\gamma=\dfrac{1}{3}$, from which we deduce

$$(a,b,c)=\left(\frac{\sqrt{2}}{2},\sqrt{2},\frac{\sqrt{2}}{4}\right).$$

14. Use $(2a)^2+(3b)^2+12ab-c^2=0$

$\Rightarrow(2a+3b-c)(2a+3b+c)=0$

15. Conceptual

16. (P) $\displaystyle\lim_{x\to1}\int_1^{x^2}\frac{f(t)-t}{(x-1)^2}dt=\lim_{x\to1}\frac{f(x^2)-x^2+2x^2f'(x^2)-2x^2}{1}$

$$=\lim_{x\to1}\frac{2x\left(\left(f(x^2)\right)-x^2\right)}{2(x-1)}=\lim_{x\to1}\frac{f(x^2)-x^2+2x^2f'(x^2)-2x^2}{1}=4$$

(Q) $\displaystyle\lim_{n\to\infty}\left(1+\sqrt[n]{4}\right)^n=\lim_{n\to\infty}\left(1+\frac{\sqrt[n]{4}-1}{2}\right)^n$

$$=e^{\lim\limits_{n\to\infty}\left(\frac{4^{1/n}-1}{2}\right)n}\qquad=e^{\left(\lim\limits_{n\to\infty}\frac{1}{2}\left(\frac{4^{1/n}-1}{1/n}\right)\right)}=e^{\frac{1}{2}.n4}=2$$

(R) $f(x)=\displaystyle\lim_{n\to\infty}\frac{2x}{\pi}\tan^{-1}(nx)=x;x>0$

$\displaystyle\lim_{x\to0^+}[f(x)-1]=\lim_{x\to0^+}[x-1]=-1$ (S) $\displaystyle\lim_{n\to\infty}\left[\sum_{r-1}^{n}\frac{1}{2^r}\right]=\lim_{n\to\infty}\left[1-\frac{1}{2^r}\right]=-0.$

17. (A) 2 sinx, sin 2x & 2 cosx are in A.P., hence sin 2x = sin x + cos x

Now, let $\sin x + \cos x = t,$ then $\sin 2x = t^2 - 1,$ then the above equation gives

$t^2 - t - 1 = 0$ or $t = \sin x + \cos x = \dfrac{1-\sqrt{5}}{2}$

(B) Let C be the sides angles, then angles will be $\dfrac{\pi}{3} - \dfrac{c}{3}, \dfrac{\pi}{3}, \dfrac{\pi}{3} + \dfrac{c}{3}$

$\Rightarrow \dfrac{\pi}{3} + \dfrac{c}{3} = c \Rightarrow c = \dfrac{\pi}{2},$ hence angles are $\dfrac{\pi}{6}, \dfrac{\pi}{3}, \dfrac{\pi}{2}$

Now $\dfrac{a}{\sin\frac{\pi}{6}} = \dfrac{b}{\sin\frac{\pi}{2}} = \dfrac{c}{\sin\frac{\pi}{2}} \Rightarrow \dfrac{a}{1} = \dfrac{b}{\sqrt{3}} = \dfrac{c}{2}$

Hence angles are $\dfrac{\pi}{6}, \dfrac{\pi}{3}, \dfrac{\pi}{2}$

Now $\dfrac{a}{\sin\frac{\pi}{6}} = \dfrac{b}{\sin\frac{\pi}{2}} = \dfrac{c}{\sin\frac{\pi}{2}}$ $\qquad \dfrac{a}{1} = \dfrac{b}{\sqrt{3}} = \dfrac{c}{2}$

(C) Let $AB = a, BC = \sqrt{3}\,a, CA = 3a$ & $AD = 3\sqrt{3}\,a$

Now $\cos A + \cos C = 0 \Rightarrow \dfrac{a^2 + 27a^2 - BD^2}{3\sqrt{3}\,a^2} + \dfrac{a^2 + 9a^2 - BD^2}{3\sqrt{3}\,a^2} = 0$

$BD^2 = 20a^2 \Rightarrow \dfrac{BD}{AB} = \sqrt{20} = 2\sqrt{5}$

(D) $a + c = 2b \,\&\, b^2 = \dfrac{2a^2c^2}{a^2 + c^2} \Rightarrow (a+c)^2 = \dfrac{8a^2c^2}{a^2 + c^2}$

$\Rightarrow (a+c)^2 - 2ac(a+c)^2 - 8a^2c^2 = 0 \dfrac{(a+c)^2}{ac} = -2, 4$

But gives $a = c,$ hence taking – 2 $\qquad a^2 + 4ac + c^2 = 0 \Rightarrow \dfrac{a}{c} = \sqrt{3} - 2.$

PHYSICS

18. $\vec{a}_0 = \left(2g\hat{i} + g\hat{j}\right) \quad \vec{F}_B = \rho V\left(\vec{a}_0 - \vec{g}\right) = 2\rho Vg\left(\hat{i} + \hat{j}\right)$

$\vec{F}_{net} = \vec{F}_B + m\vec{g} = 2\rho Vg\left(\hat{i} + \hat{j}\right) - \rho_0 Vg\hat{j}$

$\vec{a}_{net} = \dfrac{2\rho}{\rho_0} g\left(\hat{i} + \hat{j}\right) - g\hat{j} \qquad \vec{a}_{relative} = \vec{a}_{net} - \vec{a}_0$

19. Most probable velocity $V_{mp}\,\alpha\sqrt{\dfrac{T}{M}}$

From graph, $V_1 < V_2 \Rightarrow \dfrac{T_1}{M_1} < \dfrac{T_2}{M_2}$

20. Applying Bernaullis theorem from the frame of car. $P_{in} + 0 = P_{out} + \dfrac{1}{2}\rho v^2$

Again $\left(P_{in} - P_{out}\right)\times\left(0.4\right)^2 = \sigma \times 1.6 \times t$

$\Rightarrow \dfrac{1}{2}\rho v^2 \times 0.16 = 4\times 10^7 \times 1.6t \quad \Rightarrow \dfrac{1}{2}\times\dfrac{1.25\times 40\times 40\times 0.16}{1.60\times 4\times 10^7} = t \qquad t = 2.5\,\mu m$

21. $a_Q > a_P \qquad \dfrac{3g\,r\sin\theta}{l} > \dfrac{4}{3}g\sin\theta \Rightarrow \quad r > \dfrac{4l}{9}$

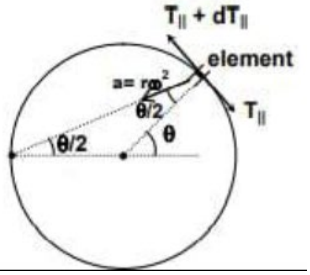
22. $\vec{a}_B = \vec{a}_A + \left(\vec{\alpha}\times\overrightarrow{r_{B,A}}\right) - \omega^2\overrightarrow{r_{B,A}} \qquad = -5j + 4\alpha i + 3\alpha j - \left(12i - 16j\right)$

$\vec{a}_B$ along y-axis should be zero.

$\Rightarrow 11 + 3\alpha = 0 \quad \Rightarrow \alpha = -\dfrac{4}{3} rad/s^2$

$a_B = \left(4\alpha - 12\right)i = -\dfrac{80}{3} i\, m/s^2$

23. $dT_{\parallel} = \left(dm\right)r\omega^2 \sin\dfrac{\theta}{2} \Rightarrow T = T_A + \dfrac{MR\omega^2}{2\pi}\left(1 - \cos\theta\right)$



24. Given 10 VSD = 9 MSD

Least count = $\dfrac{1MSD}{10} = 0.1mm$

As '0' of V.S. lies before '0' of M.S.

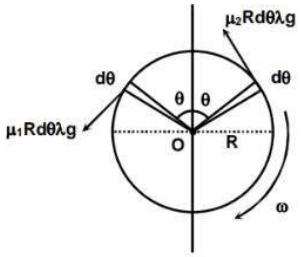
Zero error = –[10 – 6] L.C.

= – 4 × 0.01 cm = –0.04 cm

Reading = 4.1 cm + 2 × LC
= 4.1 cm + 2 × 0.01 cm = 4.12 cm
True diameter = Reading – Zero error = 4.12 – (–0.04) cm = 4.16 cm

25. $\frac{ml^2}{12}\alpha = -\left\{\frac{2k\theta}{12}+\frac{k\theta}{4}\right\}l^2 \Rightarrow \alpha = -\left(\frac{5k}{m}\right)\theta$

26. $f = 2\left[\int_0^{\pi/2}(\mu_1-\mu_2)Rg\lambda\sin\theta d\theta\right] = 2(\mu_1-\mu_2)R\lambda g \therefore \quad a = \frac{2(\mu_1-\mu_2)R\lambda g}{2\pi R\lambda} = 4$

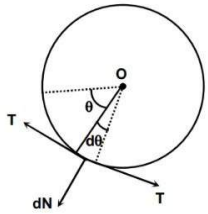


27. $2T\sin\frac{d\theta}{2} - dN = \lambda R d\theta \frac{v^2}{R}$ (λ is linear mass density of belt)

$\therefore \quad Td\theta - dN = \lambda v^2 d\theta$

$\therefore \quad$ so total normal force $= \int_{-\frac{\pi}{2}}^{+\frac{\pi}{2}} dN\cos\theta = \int_{-\frac{\pi}{2}}^{+\frac{\pi}{2}} T\cos\theta d\theta - \lambda v^2 \int_{-\frac{\pi}{2}}^{+\frac{\pi}{2}} \cos\theta d\theta$

$\therefore N = 8 newton$

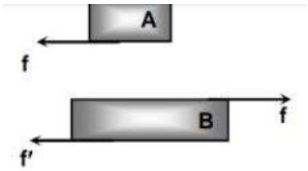


28. Since $\frac{\mu}{4} < \frac{\mu}{2} < \mu,$ the upper block will move faster than the middle block and hence force of friction on upper block is towards left

$\therefore \quad f = \frac{\mu}{4}mg \qquad \therefore \quad a_A = \frac{\mu g}{4}$ leftward

$f' = \frac{\mu}{2}.\frac{3m}{2}g \qquad a_B = \frac{\frac{3}{4}\mu mg - \frac{1}{4}\mu mg}{\frac{m}{2}} = \mu g$ leftwards

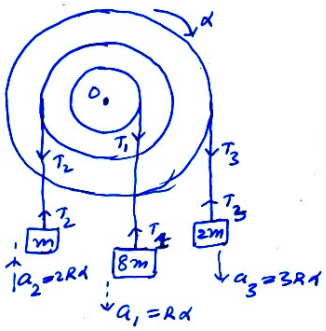
$a_{A/B} = \mu g - \frac{\mu g}{4} = \frac{3}{4}\mu g \quad \frac{3L}{8} = \frac{1}{2}\left(\frac{3}{4}\mu g\right)t^2 \qquad \Rightarrow t = \sqrt{\frac{L}{\mu g}} = 3s$



29. Acceleration of particle with respect box is along perpendicular to inclined plane

$$v_{\min} = \sqrt{g\cos 37°\left(AB + CD + AC\right)} = \sqrt{8\left(2+3+3\right)} = 8m/s$$

30. A system of pullies is massless, $\left(\tau_{net}\right)_0 = 0$



$$T_1R + T_3\left(3R\right) - T_2\left(2R\right) = 0$$

$$T_1 + 3T_3 = 2T_2 \quad _____ (1)$$

$$8mg - T_1 = 8m\left(R\alpha\right)$$

$$T_1 = 8m\left(g - R\alpha\right) \quad _____ (2)$$

Similarly,

$$T_3 = 2m\left(g - 3R\alpha\right) \quad _____ (3)$$

$$T_2 = m\left(g + 2R\alpha\right) \quad _____ (4)$$

Substituting (2), (3), (4) in (1)

$$8m\left(g - R\alpha\right) + 6m\left(g - 3R\alpha\right) = 2m\left(g + 2R\alpha\right)$$

$$12g = 30R\alpha \Rightarrow R\alpha = \frac{12g}{30} = \frac{120}{30} = 4\,ms^{-2}$$

31. $$\left(P\right)\tan\theta = \frac{1}{T\alpha} \Rightarrow 3 = \frac{VdT}{TdV} \Rightarrow \frac{3dV}{V} = \frac{dT}{T}$$

$$\ln V^3 = \ln T + C \Rightarrow TV^{-3} = \text{constant}$$

This is polytropic process with $x - 1 = -3 \Rightarrow x = -2$

$$W = \frac{nR}{x-1}\left(T_1 - T_2\right) = \frac{2\left(\frac{25}{3}\right)}{-3}\left(-1\right) = +\frac{50}{9}J$$

$$\left(Q\right)\ \Delta U = nC_V\Delta T = \frac{2\left(\frac{25}{3}\right)}{\left(\frac{5}{3}-1\right)}\left(1\right) = 25J$$

$(R)\ \Delta Q_P = nC_P\Delta T = (2)\left[\frac{5R}{2}\right]\Delta T = 5\times\frac{25}{3}(1) = \frac{125}{3}J$

$(S)\ \Delta U = 25J$ for any polytropic co-efficient i.e., for any value of 'θ'.

32. $(P)\ t = \frac{3a}{2V} = \frac{2}{3}s$

$(Q)\ \omega_0 = \frac{(V_\perp)_{rel}}{x} = \frac{\frac{\sqrt{3}V}{2}}{a} = \frac{\sqrt{3}}{2}$

$(R)\ \omega_t = \frac{(V_\perp)_{rel}}{x} = \frac{\sqrt{3}V}{2x}$

$\alpha = \frac{d\omega}{dt} = \frac{\sqrt{3}V}{2}\left(-\frac{1}{x^2}\right)\left(\frac{dx}{dt}\right) = \frac{\sqrt{3}V}{2x^2}.\left(\frac{3V}{2}\right)$ $\qquad \alpha_0 = \frac{3\sqrt{3}}{4}$

$(S)\ x = a - \frac{3V}{2}t = a - \frac{3V}{2}\left(\frac{a}{4V}\right) = \frac{5a}{8}$

$A = \frac{1}{2}(b)(h) = \frac{1}{2}(x)\left(\frac{\sqrt{3}x}{2}\right)$ $\qquad = \frac{\sqrt{3}}{4}\left[\frac{5a}{8}\right]^2 = \frac{25\sqrt{3}}{256}$

33. A) If $\frac{m_1}{m_2}\to 0 \Rightarrow m_1 \to 0$

Then $v_{2f}\to 0$ and $v_{1f} = -v_{1i}$ $\qquad \therefore\ \frac{v_{1i}}{v_{1f}} = -1$

When $\frac{m_1}{m_2}\to\infty\ \Rightarrow m_1\to\infty$

Then $v_{2f}\to 2v_{1i}$ and $v_{1f}\to v_{1i}$ $\qquad \therefore\ \frac{v_{1f}}{v_{1i}} = 1\ \therefore$ Correct graph is (B)

B) From above discussion, the answer is – 1

C) When $\frac{m_1}{m_2}\to\infty, \frac{v_{2f}}{v_{1i}}\to 2$

D) $\frac{v_{2f}}{v_{1i}}\to 0$ if m_2 is massive $\Rightarrow when\ \frac{m_1}{m_2}\to 0$

$\frac{v_{2f}}{v_{1i}}\to 2$ if m_2 is light $\Rightarrow when\ \frac{m_1}{m_2}$ is large

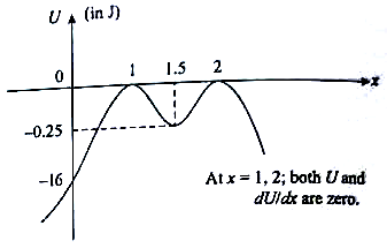
34. $U = -4(x-1)^2(x-2)^2$ $\qquad (i)$

$$\frac{dU}{dx} = -4\left[2(x-1)(x-2)^2 + 2(x-1)^2(x-2)\right]$$

Or, $\frac{dU}{dx} = -8(x-1)(x-2)[x-2+x-1] = -8(x-1)(x-2)(2x-3)$

Force on the particle is $F = -\frac{dU}{dx} = 8(x-1)(x-2)(2x-3) \qquad (ii)$

Force (i.e., $\frac{dU}{dx}$) is zero at $x = 1, 2, 1.5m.$ These are positions of equilibrium. Nature of the equilibrium can be ascertained by finding $\frac{d^2U}{dx^2}$ or just by plotting the graph of U versus x.



A) From (ii), the force on the particle is negative at $x = -1m.$ If it is released at $x = -1m,$ it will move in the negative x direction.

B) If the particle is released at $x = 1m,$ it will remain in unstable equilibrium.

C) Force $F = 8(x-1)(x-2)(2x-3)$

When the particle is displaced Δx from its equilibrium position at $x = 1.5m$ (so that its co-ordinate is $1.5 + \Delta x$), the force on it is

$$F = 8(1.5 + \Delta x - 1)(1.5 + \Delta x - 2)(3 + 2\Delta x - 3)$$

$$= 8(0.5 + \Delta x)(\Delta x - 0.5)(2\Delta x)$$

$$= 8(\Delta x^2 - 0.5^2)(2\Delta x) = 16\Delta x(\Delta x^2 - 0.5^2)$$

If $\Delta x << 0.5,$ then $\Delta x^2 - 0.5^2 \approx -0.5^2$

Or, $F = 16\Delta x(-0.5^2) = -4\Delta x$

$\therefore \quad ma = -4\Delta x \Rightarrow a = -4\Delta x \quad [\because m = 1kg]$

$\therefore$ Motion is SHM and $\omega = \sqrt{4} = 2$

D) It is same as (C)

CHEMISTRY

35. Ionic product of $D_2O\left(1.35\times10^{-15}\right)$ is less than ordinary water 1×10^{-14}

D – O bond is stronger than O – H bond so ionization of D_2O is less than H_2O

The pD value of neutral D_2O is 7.43 so if D_2O has pD 7.2 which is less than $7.43, D_2O$ is acidic. For neutral $H_2O\ pH=7$ but for D_2O is 7.43

36. I & II are aromatic.
Basic order IV > III > II > I

37. Addition of Br_2 on an alkene is a trans-addition. Q_1 is a binary mixture and is optically inactive due to external compensation. Note that the alkene P formed $\left(MeCH=CHEt\right)$ is an unsymmetrical alkene and hence anti addition of Br_2 upon it will not yield meso-product.

38. – 186 and – 104 are less than – 100°C. So they exist as liquids 60and 10 are greater than – 100. So they vapourse and do not exist as liquids.

39. C_{60} soluble in benzene. The remaining statements are incorrect

40. In version temperature $T_i=\frac{2a}{Rb}$, Boyle's temperature $T_B=\frac{a}{Rb}=T_i=2\times T_B$

Critical temperature $T_C=\frac{8a}{27Rb}$, Inversion temperature $T_i=\frac{2a}{Rb}$

Critical temperature is less than Inversion temperature
Inversion tempature more than critical temperature and Inversion temperature.

41. $G=H-TS$

$G=U+PV-TS$

$\Delta G=\Delta U+P\Delta V+V\Delta P-T\Delta S-S\Delta T$

From the first and second laws,

$T\Delta S=\Delta U+P\Delta V$

$\therefore \quad \Delta G=V\Delta P-S\Delta T$

$G=H+T\frac{\Delta G}{\Delta T}$

OR

$G=H+T\left(\frac{\partial G}{\partial T}\right)_P$

$-\frac{H}{T^2}=-\frac{G}{T^2}+\frac{1}{T}\left(\frac{\partial G}{\partial T}\right)_P$

$=\left[\frac{\partial\left(G/T\right)}{\partial T}\right]_P$

$H=-T^2\left[\frac{\partial\left(G/T\right)}{\partial T}\right]_P$]

42. $\left[Be(CH)_3\right]_n, \left[BeH_2\right]_n, \left(AlH_3\right)_n, Al_2(CH_3)_6$, $B_2H_6, Be(BH_4)_2$ and $Al(BH_4)_3$ contain 3c-2e bonding $\left[BeCl_2\right]_n, Al_2Cl_6$ contain 3c-4e bonding

43. $O^{2-}, O_2^{2-}, O_2^{-}, O_3^{-}$ ions hydrolyse in water liberating oxygen converting into hydroxide ions N^{3-} gives NH_3, C_2^{2-} gives C_2H_2, C_3^{4-} gives propyne by reacing with water. $F^{-}, Cl^{-}, OH^{-}, SO_4^{2-}$ and NO_3^{-} ions exists in water as hydrated ions

44. $\frac{K_{P1}}{K_{P2}} = \frac{4P_1}{P_2} = \frac{4\times1}{36} = \frac{1}{9} \Rightarrow K_{P_2} = 9K_{p1}$

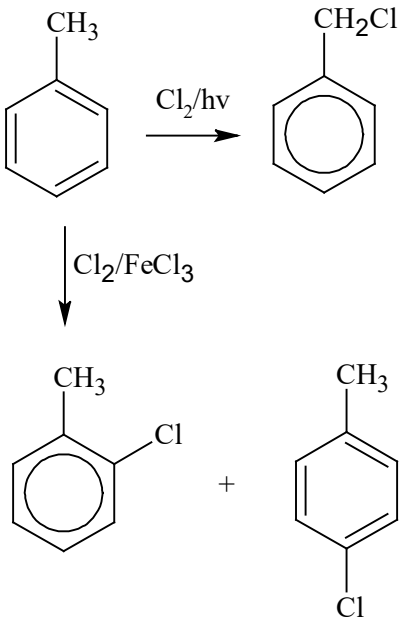
45. Given $\lambda_2 = 30.4\times10^{-7}\,\text{cm}$ $\lambda_1 = 108.5\times10^{-7}\,\text{cm}$

let excited state of He^{+} be n_2. it comes from n_2 to n_1 and then n_1 to 1 to emit two successive photon $\frac{1}{\lambda_2} = R_H.z^2\left(\frac{1}{1^2} - \frac{1}{n_1 2}\right)$ $\frac{1}{30.4\times10^{-7}} = 109678\times4\left[\frac{1}{1^2} - \frac{1}{n_1^2}\right]$ $n_1 = 2$

Now $\frac{1}{\lambda_1} = R_H z^2\left[\frac{1}{2^2} - \frac{1}{n_2^2}\right]$ $\frac{1}{108.5\times10^{-7}} = 109678\times4\left[\frac{1}{2^2} - \frac{1}{n_2^2}\right]$

$n_2 = 5$ thus excited state for He is 5th orbit.

46. x is



47. Each aromatic ring can have three resonating forms.

48.

49. **Mixture I:**

End point with phenolphthalein (disappearance of pink colour) corresponds to the neutralisation of $NaOH$ and half- neutralisation of Na_2CO_3.

$NaOH + HCl \rightarrow NaCl + H_2O$

$Na_2CO_3 + HCl \rightarrow NaHCO_3 + NaCl$

End point with Methyl orange (Appearance of Red colour) corresponds to the neutralisation of $NaOH$ and Na_2CO_3.

$NaOH + HCl \rightarrow NaCl + H_2O$

$Na_2CO_3 + 2HCl \rightarrow 2NaCl + CO_2 + H_2O$

Volume of HCl required for neutralisation of Na_2CO_3

$= 2(x-w)$

Normality of $Na_2CO_3 = \dfrac{1\times 2(x-w)}{100}$

$2(x-w)\times 10^{-2}$

Molarity of $Na_2CO_3 = (x-w)\times 10^{-2}$

Volume of HCl req. for neutralisation of $NaOH$

$= w-(x-w)$

$= (2w-x)ml$

Hence, molarity of $NaOH = \dfrac{(2w-x)\times 1}{100}$

$= (2w-x)\times 10^{-2}M$

Mixture II :

End point with phenolphthalein corresponds to half- neutralisation of Na_2CO_3 as

$Na_2CO_3 + HCl \rightarrow NaHCO_3 + NaCl$

Volume of HCl req. for complete neutralisation of Na_2CO_3 = '2y' ml

$\therefore Molarity\ of\ Na_2CO_3 = \dfrac{1}{2}\times\dfrac{2y}{100}$

$= y\times 10^{-2}$

End point with Methyl orange corresponds to neutralisation of $NaHCO_3$

Hence, volume required for neutralisation of $NaHCO_3$ present initially = $(z-2y)ml$

$\therefore$ Molarity of $NaHCO_3 = \dfrac{(z-2y)}{100}$

$= (z-2y)\times 10^{-2}$

50. Conceptual

51. Conceptual